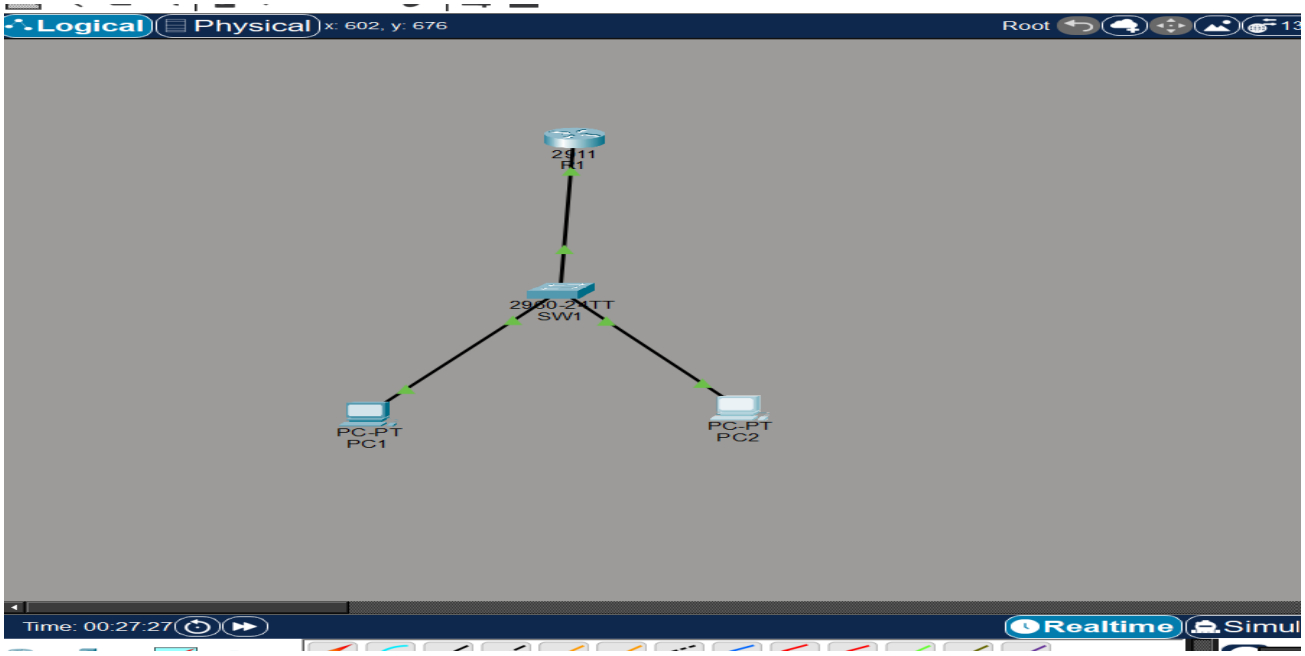
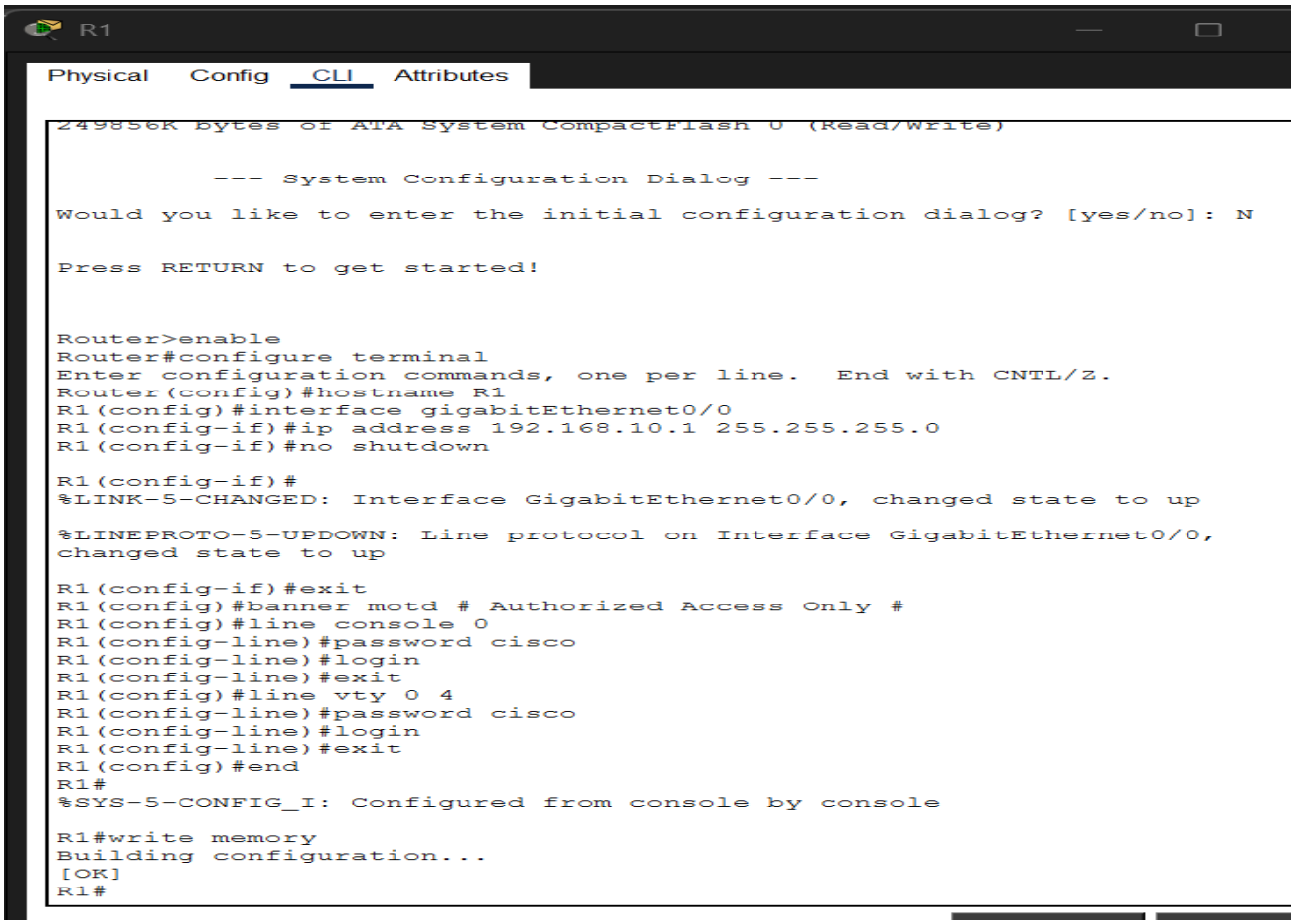
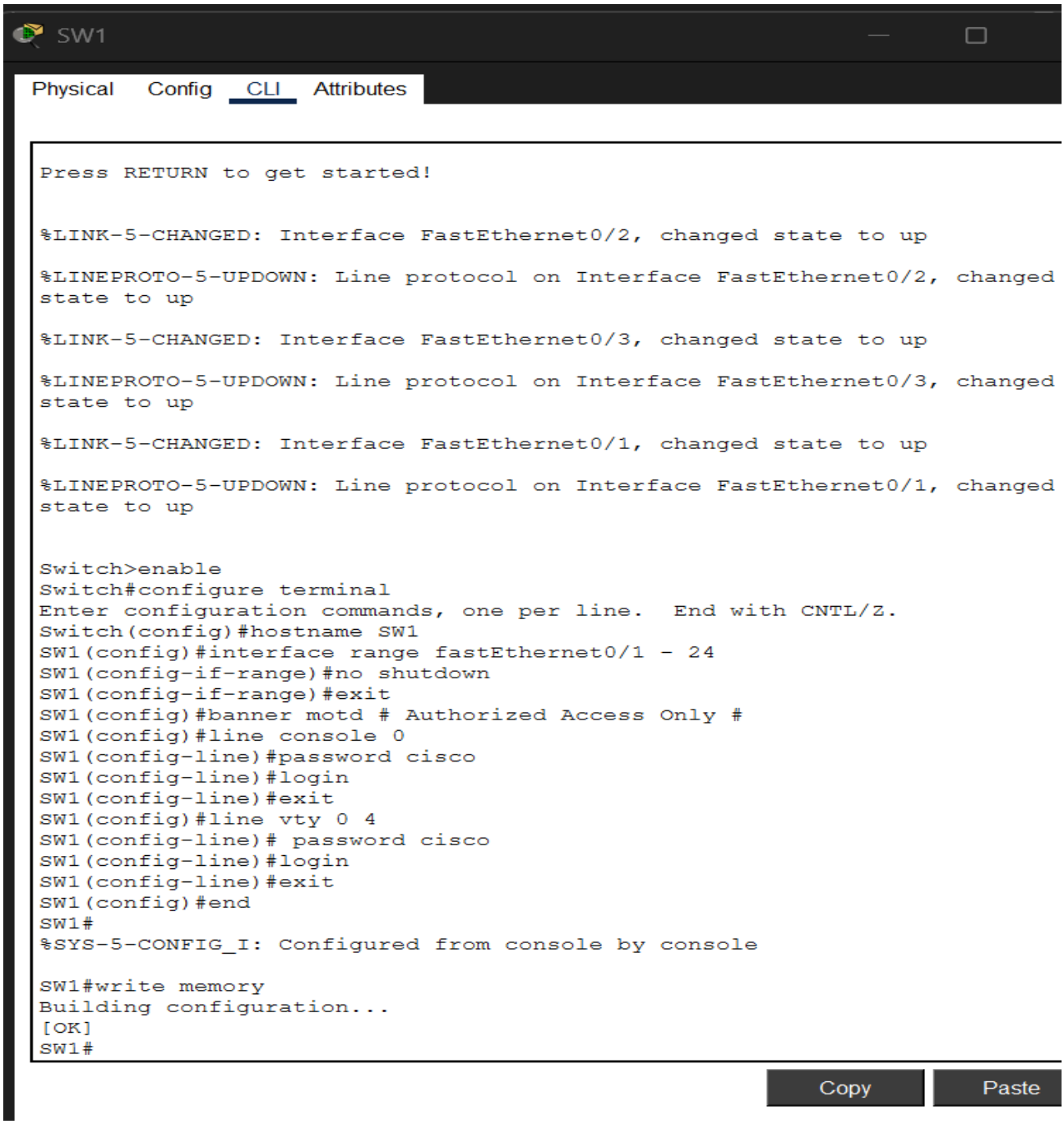
Week 4: The Cisco Troubleshooting Methodology







Deliberate Fault

PC2 has been configured with an incorrect default gateway:

Plain Text

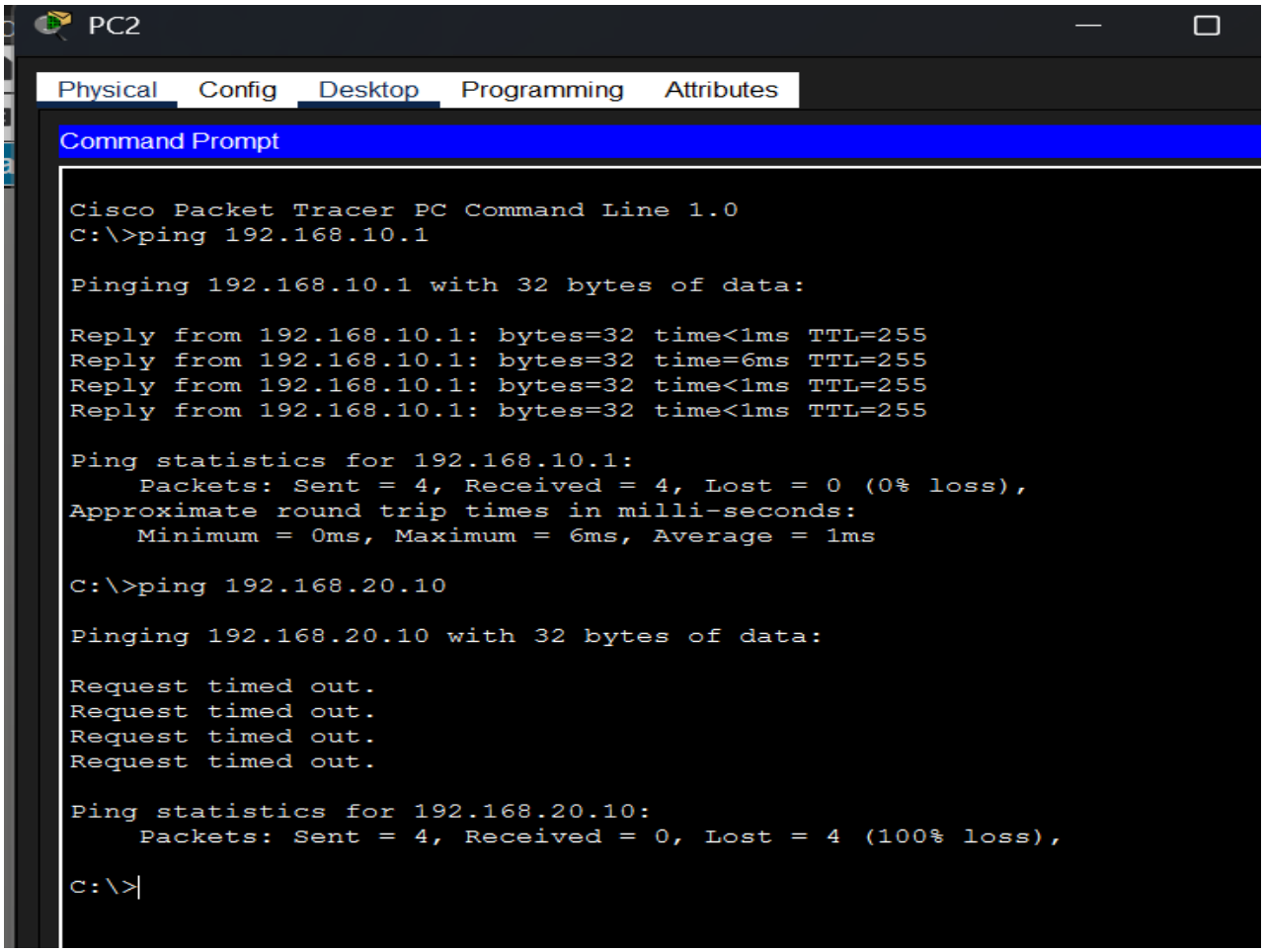
Configured Gateway: 192.168.1.1

Correct Gateway: 192.168.10.1

Effect

- PC1 can communicate with the router.
- PC1 and PC2 can communicate on the same LAN.
- PC2 will experience issues reaching networks outside its subnet because traffic is sent to a non-existent gateway.
- The fault is intentional and should remain documented as part of the topology.

1. Attempt to reach the DNS server from a client device and confirm it fails.



Conclusion

DNS server reachability test performed.

Connectivity to the DNS server failed as expected.

The failure is caused by the **intentional default gateway misconfiguration on PC2**, which is the documented fault in the topology.

1. Work through the OSI layers systematically (Physical → Data Link → Network → Transport) to isolate where the fault lies. Document the checks you perform and the results at each layer.

Layer 1: Physical Layer

Objective

Verify that physical connectivity exists between devices.

Checks Performed

PC2 Network Interface

- Confirmed PC2 is powered on.
- Confirmed FastEthernet interface is enabled.

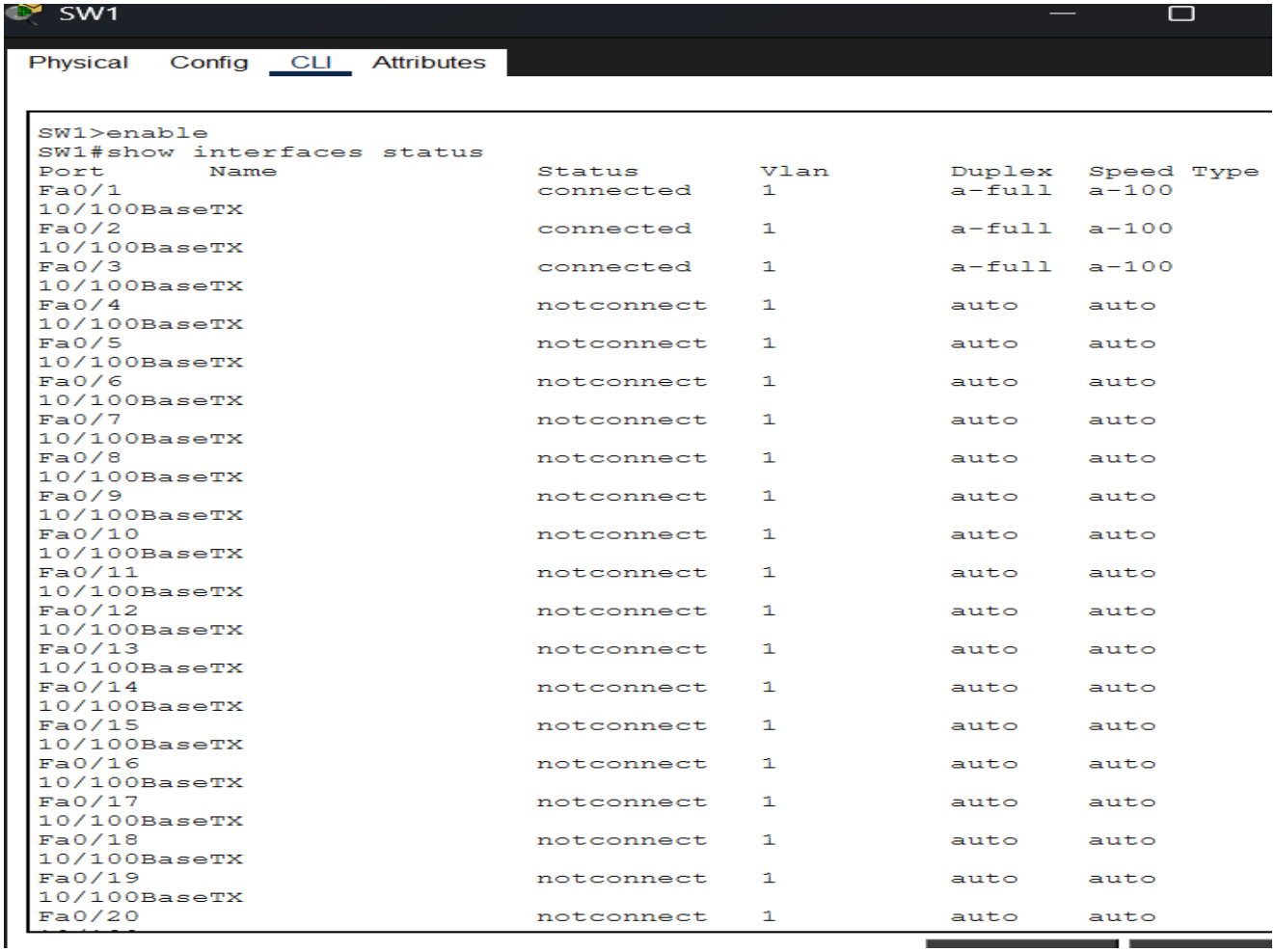
Cabling

- Verified copper straight-through cable connects:
  - PC2 → SW1 Fa0/3
  - SW1 Fa0/1 → R1 G0/0

Link Status Indicators

- Checked link LEDs on PC2, SW1, and R1.
- Verified interfaces show active connections.

Switch Verification Results



Conclusion

Physical connectivity is working correctly.

No Layer 1 fault detected.

Layer 2: Data Link Layer

Objective

Verify switching and MAC address communication.

Checks Performed

View MAC Address Table

```
10/100/1000BaseTX

SW1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address         Type         Ports
----    -----------         --------     -----

   1    0003.e453.5b01      DYNAMIC      Fa0/1
SW1#
```

Verify PC-to-PC Communication

From PC2

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

ARP Verification

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.10.1          0003.e453.5b01        dynamic
  192.168.10.10         0007.ecd5.e416        dynamic

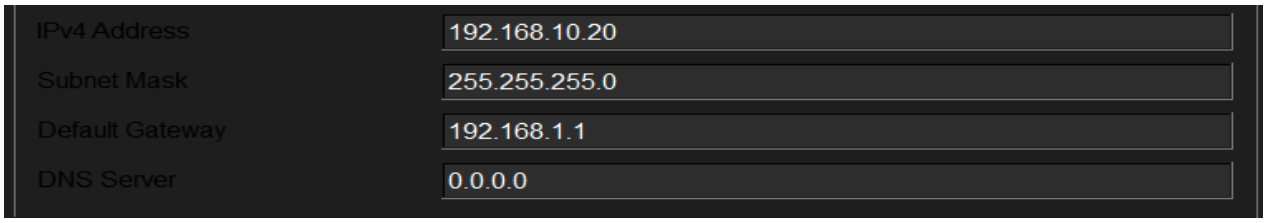
C:\>
```

Conclusion

Switch forwarding is functioning.
MAC address learning is functioning.
Devices on the same LAN communicate successfully.
No Layer 2 fault detected.

Layer 3: Network Layer

Objective
Verify IP addressing and routing configuration.
Checks Performed
Verify PC2 IP Configuration



Verify Router Interface

```
R1>enable
R1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up
up
GigabitEthernet0/1     unassigned      YES unset  administratively down
down
GigabitEthernet0/2     unassigned      YES unset  administratively down
down
Vlan1                  unassigned      YES unset  administratively down
down
R1#
```

Test DNS Server Reachability
From PC2

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Compare Gateway Addresses

Configured on PC2:
Plain Text
192.168.1.1
Router Interface:
Plain Text
192.168.10.1
Finding

The configured default gateway does not match the router's actual address.

Conclusion

Layer 3 fault identified.
PC2 is using an incorrect default gateway.
Traffic destined for remote networks cannot be forwarded correctly.
Root cause isolated at the Network Layer
Layer 4: Transport Layer

Objective
Determine whether TCP/UDP communication is failing.
Checks Performed
Because Layer 3 connectivity to remote networks fails, Transport Layer sessions cannot be established.
Conclusion

Transport Layer services are unavailable.
However, the Transport Layer is not the root cause.
The failure is inherited from the Layer 3 addressing problem.
Root Cause
PC2 has an incorrect default gateway configured.
Plain Text
Configured Gateway: 192.168.1.1
Correct Gateway: 192.168.10.1
The fault was isolated at **OSI Layer 3 (Network Layer)** after Layers 1 and 2 were verified to be functioning normally. The fault remains documented and uncorrected as required.
1. Root Cause of the Connectivity Failure
After troubleshooting through the OSI layers:

- **Layer 1 (Physical):** Passed
- **Layer 2 (Data Link):** Passed
- **Layer 3 (Network):** Failed
- **Layer 4 (Transport):** Impacted by the Layer 3 failure

Root Cause
PC2 is configured with an incorrect default gateway:

Plain Text

Configured Gateway: 192.168.1.1

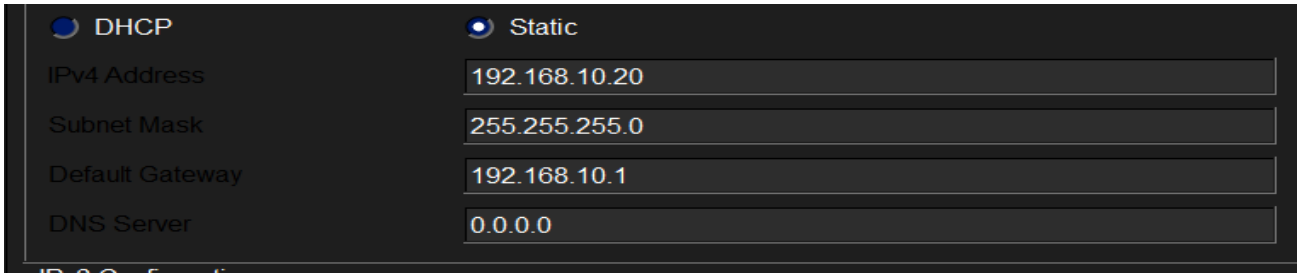
The router interface connected to the LAN is:

Plain Text

192.168.10.1

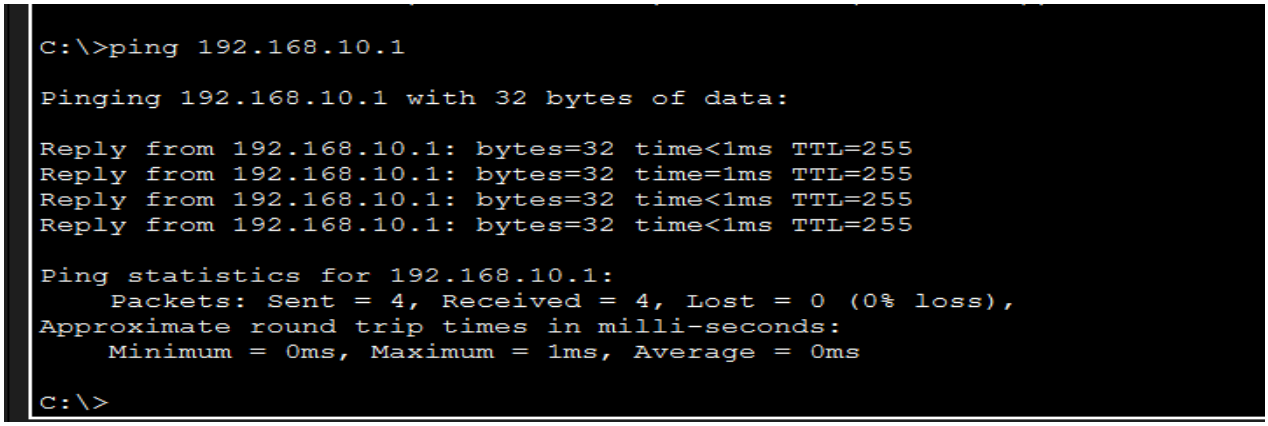
Because the default gateway does not exist on PC2's subnet, traffic destined for the DNS server cannot be routed beyond the local network.

Corrected Configuration



3. Re-Verify Connectivity

Step 1: Verify Reachability to Router



Verification Summary

Test	Before Fix	After Fix
PC2 → Router	Success	Success
PC2 → DNS Server	Failed	Success
DNS Name Resolution	Failed	Success

Conclusion

The connectivity failure was caused by an **incorrect default gateway configured on PC2**. After changing the gateway from **192.168.1.1** to **192.168.10.1**, connectivity to the DNS server was restored and DNS services operated normally. The fault has been successfully resolved and verified.

Reflection Question 1

At which OSI layer was the fault, and what evidence led you to that conclusion?

The fault was located at **Layer 3 (Network Layer)** of the OSI model.

The evidence leading to this conclusion was gathered through systematic troubleshooting:

- **Layer 1 (Physical Layer)** was functioning correctly because all devices were powered on, the cables were properly connected, and the switch and router interfaces showed an active link status.
- **Layer 2 (Data Link Layer)** was functioning correctly because the switch successfully learned MAC addresses, and devices on the same LAN could communicate with each other.
- During **Layer 3 (Network Layer)** testing, PC2 was found to have an incorrect default gateway configured (**192.168.1.1**) instead of the router's actual gateway address (**192.168.10.1**).
- Although PC2 could communicate with devices on its local subnet, it could not reach the DNS server on a different network because packets intended for remote destinations were being sent to an invalid gateway.
- Once the default gateway was corrected to **192.168.10.1**, connectivity to the DNS server was successfully restored.

Conclusion: The fault was a **Network Layer (Layer 3) configuration error** caused by an incorrect default gateway on PC2. The inability to reach networks beyond the local subnet, combined with successful Layer 1 and Layer 2 tests, confirmed that the root cause was at Layer 3.